

Calibrated Perplexity for Robust AI-Generated Text Detection

Vladimir Chaikin
Moscow Institute of Physics and Technology
vladimir.chaikin@phystech.edu

Course: My First Scientific Paper
Expert: A. Grabovoy | Consultant: A. Voznyuk

Problem: Domain Shift in Perplexity

Same text, different perplexity across domains.

News article: PPL = 11.7 Scientific paper: PPL = 27.3

A single threshold fails across different genres. The perplexity of the same text varies by 2–3× depending on which domain’s language model you use.

Research question: *How to make perplexity-based detection robust to domain shift without knowing the test text’s domain?*

Our Solution

Ensemble of Domain-Specific KenLM Models

- Train separate trigram KenLM models on **7 domains**: news, hard_essays, scientific texts, stories, articles, hard_news, hard_science
- For a new text, compute perplexity from **all models**
- Combine predictions via **weighted voting**

Key advantage: No need to know the test text’s domain — each model specializes.

Calibration Objective

We minimize the variance of thresholds across domains:

$$\sum_{i=1}^{N_d} \sum_{x_j \in X_i} (M_{\mu_i,i}(x) - y_{ij})^2 + \sum_{i < j \leq N_d} (\text{TH}_{M_{\mu_i,i}}(x) - \text{TH}_{M_{\mu_j,j}}(x))^2 \rightarrow \min_{\vec{\mu}}$$

Each model gets its own threshold:

$$\tau_d = \arg \max_{\tau} \text{F1} \left(\mathbb{1}[\text{PPL}_{\mathcal{M}_d}(x) < \tau], y(x) \right)$$

Ensemble decision:

$$H(x) = \mathbb{1} \left[\sum_{d \in \mathcal{D}} w_d \cdot \mathbb{1}[p_d < \tau_d] \geq 0.5 \right]$$

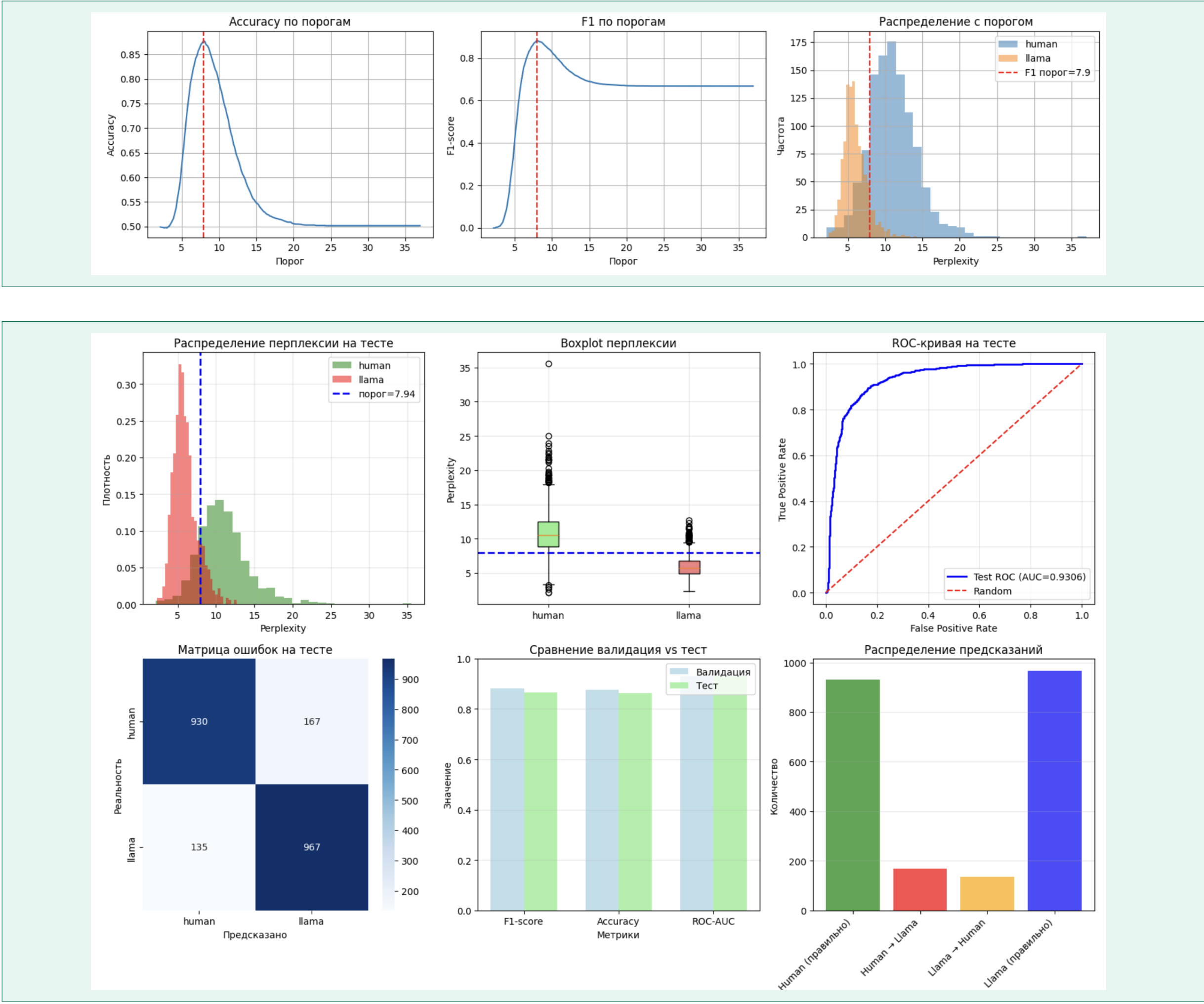
Conclusion

- Perplexity thresholds are **highly domain-dependent** (2–3× variance)
- Ensemble of domain-specific KenLM models for **domain-agnostic detection**
- Voting achieves the **most robust results**

Future Work

- Upgrade combined ensemble with neural models
- Test on more diverse domains and languages
- Cluster domains automatically
- Combine with other detectors (GLTR, Binoculars)

Experiment 1: Cross-Domain Perplexity Matrix



Train / Test	stories	reviews	articles
AINL (science)	27.3 / 23.4	26.1 / 21.9	23.8 / 19.2
stories	16.8 / 10.6	16.1 / 14.2	20.0 / 18.7
reviews	18.5 / 13.8	11.1 / 9.3	18.8 / 15.8
articles	20.1 / 15.2	17.7 / 13.6	13.3 / 9.0

Cell format: Human PPL / AI PPL. **Bold diagonal** = in-domain.

Experiment 2: Tokenization and Pruning

Goal: Improve model stability across domains

Methods tested:

- SentencePiece tokenization (vocab_size=8000)
- N-gram pruning (–prune 2 2 2)
- Size cutting
- Combined optimization

Result: Only 1–2% improvement – helpful but not sufficient.

Experiment 3: Ensemble Strategies

Method	Human recall	AI recall
Simple majority	81%	71%
Perplexity-weighted	82%	70%
Percentage-weighted	81%	70%

Key insight: All ensembles show almost equal results — robust.